Date: / /

Experiment No. 18

Aim: To determine the boiling point of an organic compound.

Apparatus: Thiele's tube, retort stand with clamps thermometer, capillary tube, fusion tube, thread, burner, etc.

Chemicals: Given organic liquid compound, liquid paraffin.

Theory: Boiling point of a liquid, is the temperature at which vapor pressure of a liquid is equal to the external atmospheric pressure. (Boiling point changes with change in external pressure.)

Procedure:

1. Fill small quantity of given organic liquid in a fusion tube.
2. Take a capillary tube seal its one end by mean of heating and place it in a fusion tube inverted (sealed end upward and open end dipped in the liquid.)
3. Tie the fusion tube near the bulb of the thermometer by thread.
4. Now, suspend the thermometer tied with fusion tube in Thiele's tube containing liquid paraffin such that it should not touch the sides of the Thiele's tube and organic liquid in fusion tube should be completely dipped in paraffin. Precaution should be taken such that paraffin should not be enterd in fusion tube.
5. Heat gently the side arm with waving flame.
6. Stop the heating, when bubbles start coming out from the open end of the capillary tube.
7. Record the temperature when continues bubbling in fusion tube is stop and liquid rises in the capillary.

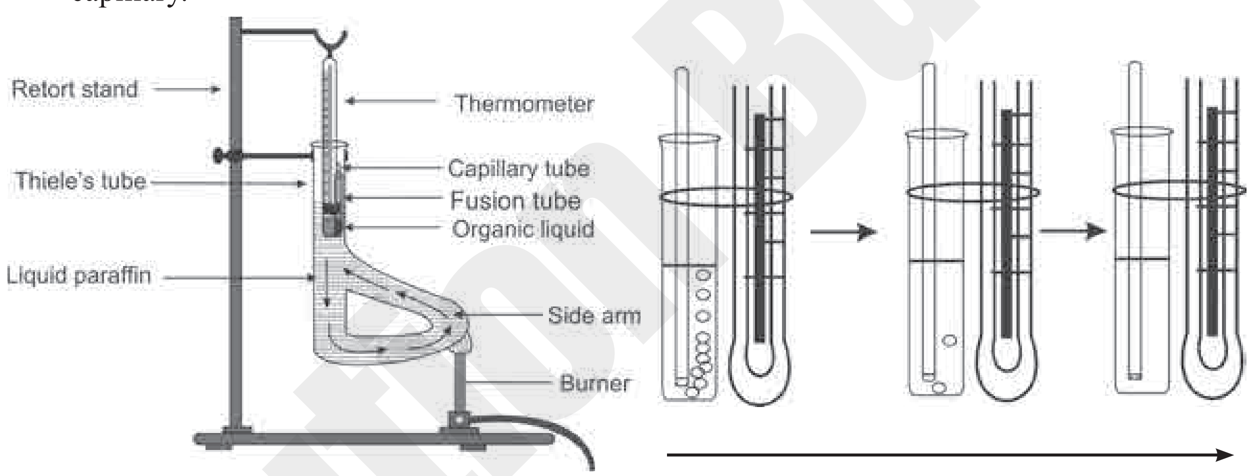


Fig. 18.1 Determination of boiling point

Observation: Boiling point of given organic compound = 56 ^{0}C

= (56 ^{0}C + 273) K = 329 K

Note: Minimum two different oraginc liquid compounds should be taken for this experiment.

Table:

Sr. No.	Compounds	Boiling Point (^{0}C)
1	Acetone	56
2	Acetic acid	118
3	Aniline	184
4	Ethyl alcohol	78

MCQ

Select [✓] or underline the most appropriate answer from given alternatives of each sub question.

1. The boiling point of a liquid is the temperature at which
   a. vapor pressure of liquid is higher than external pressure
   b. vapor pressure of liquid is lower than external pressure
   c. vapor pressure of liquid is equal to the external pressure ✓
   d. the external pressure is equal to atmospheric pressure
2. The boiling point of acetone in Kelvin is -------
   a. 273 K   c. 329 K ✓
   b. 356 K   d. 423 K
3. Which of the following apparatus is use for the determination of temperature?
   a. Thermometer ✓   c. Speedometer
   b. Barometer   d. Colorimeter
4. In which of the following stage, boiling point of organic liquid is to be recorded?
   a. At the time of heating Thiele's tube
   b. Continuous stream of bubbles start coming
   c. Last bubble is given out and liquid raise in capillary ✓
   d. When first bubble comes out of capillary

Short answer questions

1. Define the term boiling point.

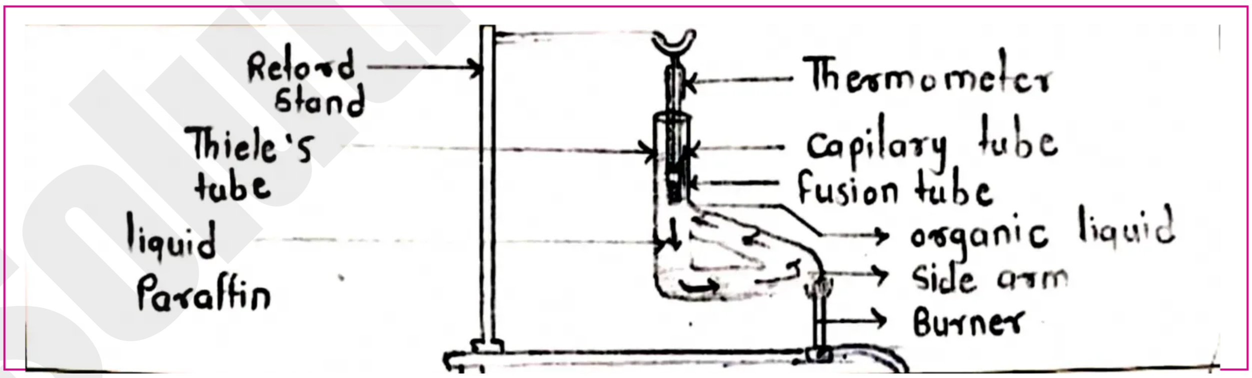
Ans. The boiling point of a liquid is the temperature at which its vapour pressure becomes equal to the external atmospheric pressure.

2. What is criteria of purity of liquid substances?

Ans. A pure liquid has a sharp and constant boiling point. Therefore, the boiling point is used as the criterion of purity of a liquid substance. If impurities are present, the boiling point changes and is not sharp.

3. Draw neat label diagram of determination of boiling point of liquid.

Ans:



Remark and sign of teacher: ....